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Embarrassingly Agile – Data Visualization Methodology in Emergency Responses

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Abstract—The pandemic had broad reaching impacts on how we do many things including the way that we design and implement visualizations. In this article, we reflect on how visualization design changed in an emergency response. Based on these reflections, we present modifications to design methodologies for visualizations to accommodate an emergency response and its working conditions.

"Speed trumps perfection. Perfection is the enemy of the good when it comes to emergency responses."
Dr Mike Ryan, March 14, 2020.

BACKGROUND

The sudden occurrence of a pandemic as a situation that must be controlled by pharmaceutical and non-pharmaceutical measures generates the need for an emergency response, in which the evolving events have to be monitored closely and intensely, and rapid decisions have to be taken on the basis of the available information. As more or different information becomes available and the first evaluation of earlier measures is performed, a different set of decisions and actions might prove to be more adequate. This process keeps iterating at a fast and largely unpredictable pace. The process of visualizing data in the course of an emergency response reflects this cycle.

It is clear that visualization is important in providing information to decision makers, but it is essential to adequately inform the public as well. The main objective of communication in a pandemic or another emergency situation is to communicate in a way that builds and maintains trust. If there is no trust in the managers of the emergency, the population might react in ways that further worsen the situation. Maintaining the public's trust in a pandemic or another serious emergency situation requires transparency, i.e., communication that is candid and easily understood. Transparency allows the public to understand the process of information gathering and decision-making [13]. Well-designed visualizations help on both fronts.

On the other hand, focusing on the developers of a visualization in such a situation, the potential psychological effects of the pandemic on the visualization designers and researchers might be very large. Those people working in the collection, analysis, visualization, and interpretation of the data are themselves individuals living in society, with their family and social networks. They or members of their immediate family could have conditions that put them at risk. They might have relatives who are seriously ill or deceased. This creates an unavoidable emotional and psychological stress that adds to the pressure of having to produce results at a very high pace in sub-ideal working conditions.

In this contribution, we reflect on our own experience and observations as researchers from multiple backgrounds who have worked on visualization during a pandemic response. We discuss here an approach and a set of strategies to adapt conventional design study methodologies—i.e., analyzing a particular domain problem, developing a suitable visualization design to solve this problem, and finally validating the

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result—to also work in emergency scenarios where time matters the most.

Although the focus of our work is on the COVID-19 pandemic, other emergency situations, as mentioned above, such as natural or man-made disasters, could have similar characteristics and would benefit from this documented experience. By putting agility and flexibility at its core, our proposed approach offers adaptability to evolving situations where data and limited resources determine which methods are feasible and effective.

CONSIDERATIONS IN AN EMERGENCY RESPONSE

From the point of view of visualization development in an emergency response situation, the key aspects that need to be accounted for are the following:

- **Big Data nature of the problem** – The four main V's of big data [17]—volume, velocity, variety, and veracity—are all present in pandemic data. Due to the criticality of the situation, large volumes of data describing various characteristics of the infection and its dynamics are presented at an unusually high velocity. Moreover, there is a large spread in their variety, since in addition to the traditional medical data, other factors such as travel, condition of the wastewater, contacts, and compliance with non-pharmaceutical measures need to be captured. The lack of sufficient time to define standard results from multiple sources in different formats affects the completeness and cleanliness of the data (veracity).
- **Resource availability** – Analyzing such data generally requires a different set of tools than those routinely present in the units that are on the front line in the immediate response. This implies that an appropriate set of hardware and development tools would need to be made available to those working in proximity to the data in a very short time frame.
- **Multidisciplinarity of the team** – People with the required and often complementary expertise need to be identified, reached out to, and brought together. The team will need to work under the guidance of a leader who has a generally high level of understanding of all the aspects of the single disciplines working symbiotically in a highly multidisciplinary project. The manager of the team can play an important role in procuring the right expertise. Therefore, the network of knowledgeable people that the manager has is an asset. It is our experience [10] that members of the team may not even know that visualization is a field, and it is important to keep this in mind.
- **Creative thinking** – While previous skills and expertise of researchers are certainly very useful, the unprecedented situation requires them to think quickly and unconventionally, going beyond frameworks and approaches they are used to. Flexibility and quick response to a fast-moving landscape are paramount.

It should be noted that this list is not exhaustive, but contains considerations based on our experience and reflections that need to be taken into account by visualization design methodologies for emergency responses.

RELATED WORK: VISUALIZATION DESIGN METHODOLOGIES

Before we consider the changes to visualization design methodologies that needed to be made in an emergency response, we should review these methodologies themselves. The seminal work of Sedlmair *et al.* [9], on which we base our described methodology, discusses a methodology for problem selection, inward-facing validation, and outward-facing validation in visualization development. The methodology has been developed through the authors' personal experience in this area. The *design study lite methodology* [11] describes a version of the above-mentioned methodology [9] that can be used in an educational setting, where time is necessarily constrained, on applied projects in the classroom with visualization students. Wood *et al.* [14] describe a methodology to design visualizations whereby the code written for a visualization best reflects the design decisions made when writing this code.

In terms of the COVID-19 response specifically, Sondag *et al.* [10] describe modifications to the design study methodology that needed to be made in response to a visualization design in a pandemic. The specific problem was a visual analytics approach to simulate policies on simulated contract tracing data. Our paper considers this work as well as our own personal experience working on visualization problems during the COVID-19 pandemic. Also, colleagues in the UK have documented the RAMPVIS pandemic response [3], [5] over the course of the pandemic itself.

In this paper, we reflect on the changes that we made to our visualization design methodologies when responding to the pandemic. Our methodology is clos-

est to and inspired by Sedlmair *et al.* [9] except some of the stages are skipped over entirely as in other methodologies [10], [11] while *Discovery*, *Design*, and *Implementation* are merged in various configurations.

Survey: Challenges of visualization projects during COVID-19 pandemic

What changes did visualization researchers make to their design methodologies in the pandemic? In February 2024, a seminar took place in Dagstuhl, Germany, with international experts in visualization, public health, and epidemiology from universities, scientific institutions, university hospitals, and health authorities. All of the participants had been involved in visualization projects during the pandemic, either on behalf of health authorities or on their own initiative. To obtain an insight into typical environmental conditions and challenges when having developed visualizations during the pandemic, but also into the procedures which were applied to deal with these conditions, a survey was conducted among these experts. The questionnaire was developed by a subgroup of participants and included seven open-ended questions. This subgroup, which compiled the questionnaire, was excluded from the survey target participants.

Ten participants of the seminar then completed the questionnaire. To summarize the essentials of the survey, the main categories were derived from the responses to the open-ended questions and the individual responses assigned to one or more categories. The results are presented in Table 1, which shows that the majority of respondents were based in academic institutions, mainly departments of visualization or computer science, but hospitals and government agencies were also represented. The respondents were mainly involved in the development of dashboards, but also in visualizing contact networks and in visualizations that support risk assessment. The latter includes visualization of risk behaviour and protection measures for the public. The time pressure in the COVID-19-related visualization projects was regarded as much higher compared to other projects. These time requirements led to increased reuse of already existing code, either from own projects or from libraries. Interestingly, a substantial subset of the respondents had no direct access to real data during the development process due to privacy and data protection issues. This was the reason for some of the teams working with artificial and simulated data. In addition, obtaining feedback from the domain experts was a serious problem for some of the respondents because those experts had none or very limited time to test and evaluate the increments of the applications and to provide feedback. Some respondents switched to requesting feedback via structured questionnaires instead of conducting regular sprint reviews, and others had to rely on unstructured user feedback.

Question / Categorized answers	*n**
1. Institution:	
Academic: visualization / computer science	7
University hospital	2
Health authority	1
2. Type of visualizations developed:	
Dashboards (maps, time series, genetics)	6
Risk assessment and communication	4
Contact networks	3
Vaccination progress	1
3. Time constraints:	
Very high time pressure	3
High time pressure	5
Conditions similar to other projects	2
4. Design methodology applied:	
Adapting existing code / applications	4
Speeding up or leaving out design stages	4
Outsourcing	2
Design as usual	1
5. Use of specific libraries:	
Own libraries	3
D3	5
JavaScript	2
Others (Node.js, R:ggplot, highcharts, leaflet)	3
6. Types of data used / accessed during development:	
Use of real data	5
Delayed or no access to real data	4
Use of simulated data	3
7. Mode of feedback:	
Jour fix / sprint review with domain experts	5
Structured feedback from domain experts	2
Unstructured feedback from user behaviour	3

TABLE 1. Results of survey among seminar participants on challenges of visualization projects during the COVID-19 pandemic and approaches to deal with these challenges. The categories were derived from the responses to open-ended questions. The total number of respondents was 10.
*: For questions 2 and 4-7, multiple categories were possible.

An interesting finding of this small survey is that 8 of 10 respondents either left out design stages or reused existing software prototypes. Other new insights are the already mentioned predominantly use of synthetic data, as well as overall difficulties in receiving expert feedback for evaluation purposes. These first results confirmed to some extent our own experiences as visualization experts/developers during the pandemic and inspired the agile data visualization methodology presented in the next section.

agile(*t*)

Based on the outcomes of the above-mentioned survey and the retrospective observations and reflections on the experience of the paper and survey authors with designing visual representations during the COVID-19 pandemic, we propose our methodology. It is inspired by the existing design study methodologies [9], [10], [11] but places the time dimension at its centre.

We introduce *agile*(*t*) as a methodology that adapts to the current situation of the evolving emergency, the availability of time and resources, and the availability of the domain experts. In an emergency situation where time availability is extremely low, we are by definition embarrassingly agile and many stages of a design study [9] are skipped. As more time is available, we enable these stages again.

In emergency situations, the first moment resembles a state that we call *embarrassingly agile* (Fig. 1, red line)—immediate unstructured reaction when the visualization experts are trying to understand the data and tasks from the experts in infectious diseases as much as possible and at the same time coming with first possible visual representations, often reusing existing tools and systems. The experts may often be less familiar with visualization and existing prototypes are often reused on existent data to help get a sense of the tasks and start a dialogue [10]. This phase lasts maximally a couple of days. Then, when time *t* for the development increases, a more systematic approach is followed. Stages continue to stabilize based on the time available—when still having limited time resources, the process can resemble agile software development (Fig. 1, purple line), or it can follow the waterfall approach (Fig. 1, green line) when there is enough time for discussions about the designs and created prototypes with the experts for infectious diseases. In this case, the iterations are converging to the final deployed solution much faster.

To identify the presence and content of individual phases of the process described in the Sedlmair *et al.* methodology [9], we looked at these phases from the perspective of available time. If we have limited time for creating visualizations as an emergency response, we have to skip several stages of their pipeline. We identified that only the most crucial stages—*Discover*, *Design*, and *Implement*—are the core ones that have to be preserved in all situations (Fig. 2). However, when the time available is very short (almost close to zero) and we are embarrassingly agile, we argue that all other stages of the design study methodology are skipped, with *Discover*, *Design*, and *Implement* merged into one, and we do them together when the domain experts are available in a single session. This corresponds to our scenario when the visualization designers, with very limited input from the experts, are trying to choose the closest appropriate representation from the existing solutions. As time increases, the Implementation stage starts to separate from the Discover and Design stages, as we start to utilize libraries and build more customized solutions. If we have even more time available, also the Discover and Design stages start to separate, and we can incorporate more and more techniques that help to better analyze the problems and choose the most appropriate solution. However, the interconnection between the stages is still crucial, as it was proposed in the original methodology. As time returns to normal, we eventually enable all stages of the methodology [9].

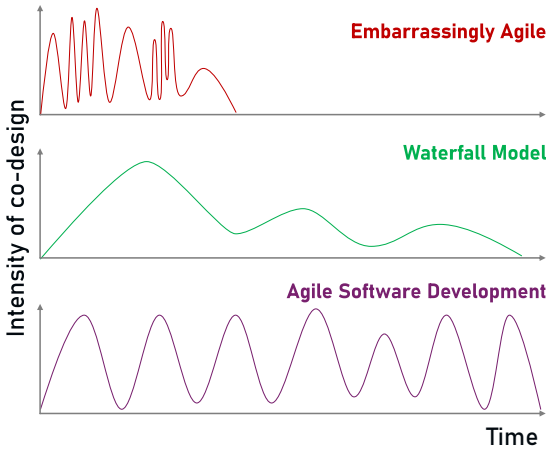


FIGURE 1. An illustrative indication of how user-engagement and co-design take place (with the experts) over the duration of a design and development process. Top: The embarrassingly agile approach during an emergency response is identified by its shorter duration and intense but unstructured engagements. Bottom: Two common approaches when more time is already available: the waterfall model and agile methodology – which are both more structured and systematic.

HOW TO BE *agile*(*t*)? – NOW

Visualization is now a mature research area where significant volumes of knowledge in the form of methods, techniques, guidelines, best practices, and empirical evidence have accumulated. It is complemented by openly available software infrastructures that help operationalize the knowledge base quickly and effectively. This offers visualization researchers sufficient grounds

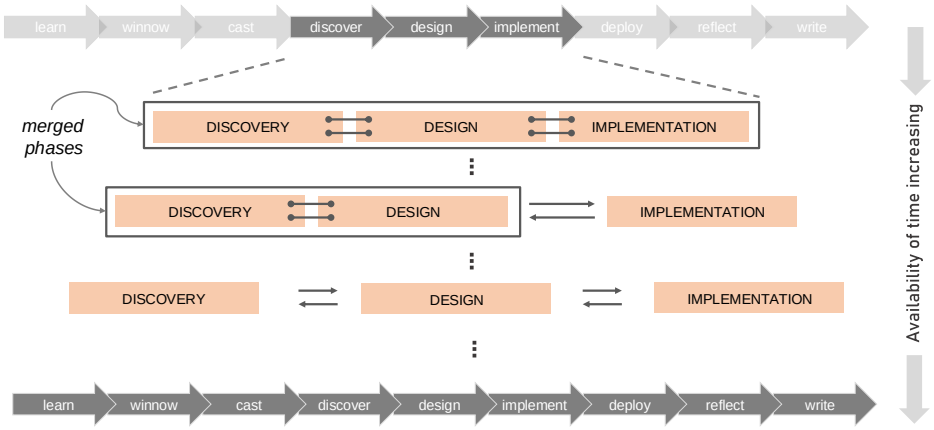


FIGURE 2. Starting from the core stages of the 9-stage design study methodology of Sedlmair et al. [9], our agile approach involves combining the *Discovery*, *Design*, and *Implementation* phases into a single activity when there is limited time. With more time available, Discovery and Design could be a single phase and be in dialogue with the Implementation phase. In cases with sufficient time and resources, all of the nine stages of DSM could be followed.

to be *agile*(*t*) already now. In this section, we present practical guidance into how visualization researchers could adopt *agile*(*t*) in their research practice by making use of the available body of knowledge existing within visualization research. The objective with the approaches we are highlighting here are all methods to respond to the key constraints of the pandemic situation – limited time and resources. Inspired by our opening quote by Dr Ryan "*perfection is the enemy of the good when it comes to emergency responses.*", we list methods that enable designers to progress *quickly* but do this *effectively* with as minimal compromises as possible in quality. Doing the design stages *quickly yet effectively* is the core mantra that underpins how *agile*(*t*) can be carried out.

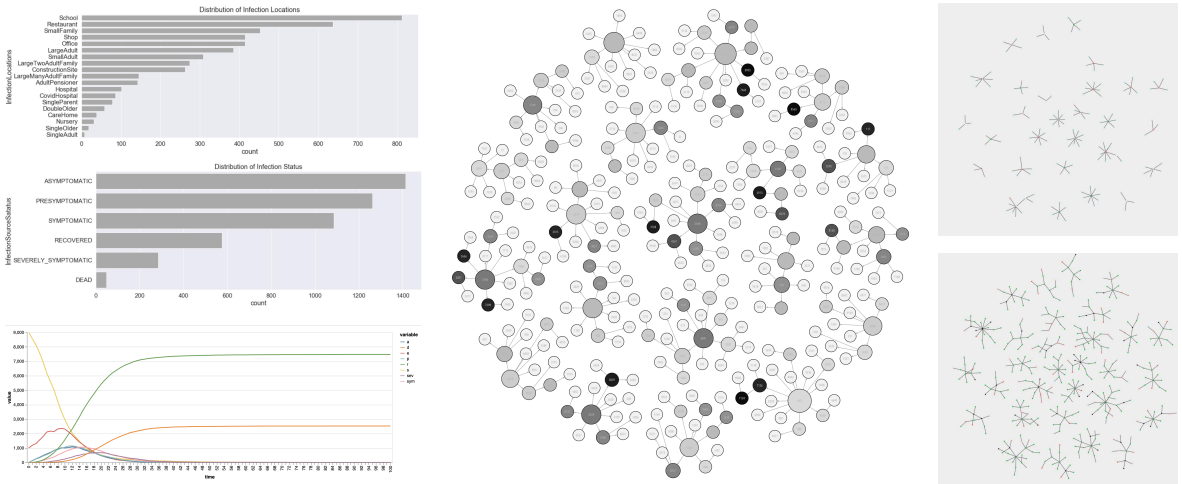
How to discover quickly yet effectively?

Performing rapid literature scanning – One of the key phases of the Discover stage is to build an understanding of the domain area and establish a shared key vocabulary. To establish a base level of understanding, visualization researchers can conduct fast literature scanning to help build an understanding of the key methods that epidemiology experts are using. Such reviews could start with key papers highlighted by the collaborators and expand as needed to build an initial picture of the problem landscape. This is akin to *rapid reviews* common in public health and medicine research [12], but what we are suggesting here is much less systematic. The rapid literature scan activity is critical for designers in building an understanding of the domain vocabulary, and also to identify existing visualization practice in the domain and to generate familiar representations to start the dialogue. For instance, the familiar line charts of key statistics from state-based epidemiological models (see an example in Figure 3-a, bottom) are one such graphic that is expected to accompany any reporting of models of this sort. When done effectively, a rapid literature scanning will surface these kinds of common visualizations.

Setting targeted yet informal discussions – While formal interviews would be ideal for gathering data for requirement and needs analysis, only brief, informal discussions will be available in emergency situations. It is therefore essential to go to these initial meetings with a basic understanding of the key vocabulary (hence the rapid review) and be upfront about key expectations from the discussions. Design teams could try and obtain insights around: key pieces of evidence needed for effective decision making, key challenges related to data and methods (such as modelling), and limitations in existing tools and software infrastructure. To more formally approach the discussions in our RAMPVIS project [3], we made use of structured diaries[1] that both document questions that to collectively explore and also use initial data visualization explorations as prompts in these discussions.

Making use of design templates – Human-centered design research offers several openly avail-

[1] https://cagatayturkay.github.io/scrc_vis/



(a) Statistical Graphics on disease spread dynamics

(b) Static graph visualisations depicting derived data

(c) Dynamic graph visualisations depicting spread over time

FIGURE 3. Using existing libraries such as Altair and Gephi and an existing research prototype [1], we created a range of visualizations of contact tracing data that we developed following an agile(t) process within one of the working groups of the RAMPVIS project [3].

able, established toolkits such as IDEO's toolkit[2] or the Service Design Toolkit repository[3] to support visualization designers to move swiftly with their design process. For instance, writing user stories for imagined users based on insights learnt from the informal discussions or making use of persona templates[4] could move groups quickly in a more informed, structured manner. Such structured models, such as user stories, were instrumental within the 30 visualization explorations[5] that we conducted within RAMPVIS.

How to design quickly yet effectively?

This next phase is concerned with getting design ideas and prototypes quickly in front of the stakeholders.

Experiment with data but make it quick and dirty – A crucial step is to understand, as Pretorius and Van Wijk put it [7], "what do the data want to be?", i.e., exploring the characteristics of the data and experimenting with representations that will emphasise these characteristics. This requires quick experiments where data can be visualized/analyzed through existing and/or off-the-shelf visualization tools and packages for building a deeper understanding of the data. Figure 3 illustrates a range of visualizations we produced within one of the working groups of the RAMPVIS project [3]. Part-a in the figure is an example of our experimentations with the contact tracing simulation data [10] that enabled the researcher team to understand what kinds of data we are working with and what potential issues might be.

Make use of existing visual vocabularies and galleries – The growing visualization research and practice communities have developed several actionable resources that distill decades of research and learning. These include collated guidelines [4], visualization galleries[6], visual vocabularies[7], and literature surveys [6] accessible through sophisticated tools [2]. An opportunistic scan of these resources and mapping them to the characteristics of the data and the needs of the stakeholders can lead to reasonable design choices that can be arrived at quickly; they are also more likely to work given all the evidence underpinning the resources. An example from the RAMPVIS collective [3] is the visual data storytelling work led by Ritsos et al.[8] that built on the narrative patterns for data storytelling (NAPA) cards[9] for their design

[2] https://www.ideo.com/journal/design-kit-the-human-centered-design-toolkit
[3] https://servicedesigntools.org/tools
[4] https://servicedesigntools.org/tools/personas
[5] https://ramp-vis.github.io/RAMPVIS-PhilTransA-Supplement/
[6] https://datavizproject.com/
[7] https://ft-interactive.github.io/visual-vocabulary/
[8] https://observablehq.com/@ritsosp/rampvis-idioms-narrative-design-patterns
[9] https://napa-cards.net/

explorations. They reflect on their report on how the existing collection of cards enabled them to structurally respond to the challenges as they emerge.

How to implement quickly yet effectively?

Implementation can often be the longest and most resource-intensive phase of developing visualization tools, but the mature state of visualization research and practice is again helpful in addressing this to a certain extent.

Re-use existing code, libraries, and platforms – The visualization software ecosystem is significantly mature at the moment. Software ranges from general-purpose visualization libraries such as D3[10] or Vegalite[11] that work also within data analysis platforms such as Jupyter, to specific tools such as Gephi[12] that can visualize graphs. Using existing software and tools not only shortens development time but is also highly instrumental in building dialogue and rapport within the researcher team [5], [3], [10]. For instance, the visuals in Figure 3 were all produced using existing tools, libraries, and research prototypes. Using tools such as Vegalite and Gephi enabled us to produce several visualizations rapidly and moved us along quickly by using the produced visuals as prompts for deeper dialogues and in several cases leading to breakthroughs within the epidemiological modelling [5]. These rapid design explorations were then the foundations of a more established design study that we carried out in the later stages of the pandemic, where we had more time and resources [10].

Visualization researcher as a "chauffeur" – In conventional design studies, the goal is to bring the tools to a deployable stage where they can be embedded in the stakeholders' workflows. In emergency situations, however, the priority is to provide effective solutions to key problems quickly. This means that compromises will need to be made in user experience and features that will enable successful deployment. An effective alternative is where the visualization researcher acts as a facilitator, a "chauffeur" of the tool in sessions where the team collectively analyses the data [10]. This reduces the impact of challenges related to onboarding and steep learning curves and gets the whole team moving more quickly towards the analytical objectives.

Conduct light-touch assessments – A fully-fledged, formal evaluation is often not possible due to the time constraints in emergency situations. Workshops, focus groups, and mediated task-led sessions [10] with the stakeholders are all effective methods that can provide valuable insights in short sessions. Heuristic evaluations[13] where designs could be tested against guidelines—such as those that we discussed above—are also effective, systematic methods for evaluation when access to users proves difficult.

A research agenda for *agile*(*t*)?

While there is already a lot of good practice to be *agile*(*t*) at the moment, there are still several open challenges and avenues for future research.

Broadening the *agile*(*t*) methodology toolkit – While this paper presents an account of existing practices that can underpin an *agile*(*t*) approach (in the previous section), these are mostly drawn from data visualization research and practice. Broadening the horizon of methods through a more systematic review of methods across data visualization, human-computer interaction, and design research needs to be conducted. A systematic mapping could look into design practices within emergency or other time-limited situations.

Building an *agile*(*t*) resource bank – As also was reported in the survey, having access to established resources to speed up the design and development phases has been critical during the COVID-19 experience. Preparing a comprehensive and easy to adopt resource that is structured by the different stages of *t* within an *agile*(*t*) process will be an impactful and useful future development. Initiatives such as VisGuides [4] set good examples of how such resources could be developed. A key open research question here is how best to package a visualization method, design approach, or resource in a way that is most easily adopted in a short amount of time. Is a templated approach the best way forward, or will that constraint innovation and diversity?

Explore *agile*(*t*) in teaching – Some of the constraints faced within emergency situations, such as the limited access to experts or limited time to conduct certain stages of a full design study, are also common in educational settings. There is potential to explore *agile*(*t*) as a method that could work within visualization classes. This is not too dissimilar to how the Design Study Lite methodology [11] has been adopted in the classroom. Given that the real pressures of a pandemic

[10] https://d3js.org/
[11] https://vega.github.io/vega-lite/
[12] https://gephi.org/
[13] https://www.nngroup.com/articles/how-to-conduct-a-heuristic-evaluation/

will not be present, these situations can be simulated by giving time-constrained tasks to students, e.g., "implement X, Y, Z within the next 10 hours and get a live version up on a webpage". Such simulated training is common in professional training such as in cyber security[14] with platforms to simulate the time and scale pressures. Future work could explore the generation of scenarios and simulated datasets with structured timelines to move students through the experience of designing within these constraints. This will build expertise within the next generation of visualization designers who are better equipped to deal with the complexities of emergency situations or other time-pressured contexts.

Methods of evaluation for *agile*(*t*) – One area that is underdeveloped within the *agile*(*t*) process is evaluation. Most of the established evaluation methods within visualization do require deep and long engagements with domain experts, which is not possible in pandemic situations. Even though we listed some light-touch assessment methods in the above section, evaluation methods that can operate effectively within an *agile*(*t*) setting and provide rigorous evidence efficiently in relation to how well a solution is working will need to be studied further. These can involve novel measurement tools or efficient online testing platforms similar to the reVISit tool[15] to streamline the evaluation process.

Provenance for *agile*(*t*) – *Provenance* has become a crucial aspect in visualization and visual analytics [15]. In both fields, it encompasses “the development of systems to visualize provenance data, analyzing such data to understand user behavior, and personalizing systems in response to user interactions” [16]. Similarly, Ragan *et al.* [8] distinguish between various provenance types, such as provenance of data, visualization, or interaction. To make informed decisions in emergency situations with the developed visualization under time pressure, we need to know where the data comes from, how it has been changed, which analysis steps (both computational and interactive actions) have been performed, as well as what the overall analytical process was performed by whom. Moreover, to become *agile(t)*, we also need provenance over the agile visualization development process, e.g., to support reusing and mixing existing visualizations. Despite the fact that a number of provenance approaches for visualization and data analysis already exist, substantially more research is needed to sufficiently support embarrassingly agile situations as described before especially when the development is done in a collaboration setting.

[14] https://www.cisa.gov/resources-tools/resources/cybersecurity-scenarios

[15] https://revisit.dev/

CONCLUSION

In the paper, we described and documented our experience developing visualizations in an emergency response and proposed a methodology based on the collective reflections of the paper authors and a survey conducted with 10 more experts at a Dagstuhl seminar. In future work, it would be interesting to collect more examples of varying levels of time pressure and how the design of visualizations was affected.

Our methodology is inspired by existing design study methodologies [9], [10], [11] but considers the important time dimension. In early stages of the emergency, only a single session involving *Discover, Design, and Implement* stages is undertaken by the experts and visualization designers together with all other stages [9] skipped. As the situation evolves, the methodology changes with *Discover, Design, and Implement* separating out and other stages enabled until it gradually becomes a carefully conducted design study [9]. We believe that this approach will be useful for many other types of emergencies and other temporally evolving design situations with time pressure.

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